

## Common types of permutations in groups

- **Identity Permutation**: A permutation that does not change the position of any element. It is denoted by  $e$  or  $id$ .
  - **Example**: In  $S_3$ , the identity permutation is  $e=(1)(2)(3)$ .
- **Transpositions**: A permutation that swaps exactly two elements while leaving all others unchanged. It is denoted as  $(ab)$ .
- **Cycles**: A permutation that cycles a group of elements in a specific order, leaving all other elements unchanged. It is written  $(a_1 a_2 \dots a_k)$ .
- **Composition of Permutations**: The operation of applying two or more permutations in sequence, denoted by  $\circ$ . For example,  $\sigma \circ \tau$  means apply  $\tau$  first, then  $\sigma$ .
  - **Example**: In  $S_3$ , if  $\sigma=(123)$  and  $\tau=(23)$ , then  $\sigma \circ \tau=(123) \circ (23)=(12)$ .

## Common types of permutations in groups(e.g.)

- **Inverse Permutations**: A permutation that reverses the mapping of a given permutation, such that  $\sigma \circ \sigma^{-1} = e$ . It is denoted as  $\sigma^{-1}$ .
- Example: In  $S_3$ , if  $\sigma = (123)$ , then  $\sigma^{-1} = (132)$ .
- Example 1: Let  $S = \{1, 2, 3\}$ ,  $\sigma = (123)$  and  $\tau = (23)$ , find the compositions  $\sigma \circ \tau$  and  $\tau \circ \sigma$ .
- Solution 1: To find the composition  $\sigma \circ \tau$ .
- ① Express  $\tau$  :  $\tau = (23) = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$ .
- ② Apply  $\sigma$  to the result of  $\tau$ , Using composition  $(\sigma \circ \tau)(x) = \sigma(\tau(x))$  :  
 $\sigma(\tau(1)) = \sigma(1) = 2$ ,  $\sigma(\tau(2)) = \sigma(3) = 1$ ,  $\sigma(\tau(3)) = \sigma(2) = 3$ .
- ③ Final Result  $\sigma \circ \tau = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} = (12)$ .

## Common types of permutations in groups(e.g.)

- Solution 2: To find the composition  $\tau \circ \sigma$ .
  - ① Express  $\sigma$  :  $\sigma = (123) = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$ .
  - ② Apply  $\tau$  to the result of  $\sigma$ : Using composition  $(\tau \circ \sigma)(x) = \tau(\sigma(x))$  :  
 $\tau(\sigma(1)) = \tau(2) = 3$ ,  $\tau(\sigma(2)) = \tau(3) = 2$ ,  $\tau(\sigma(3)) = \tau(1) = 1$ .
  - ③ Final Result  $\tau \circ \sigma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = (13)$ .
- The results of the compositions  $\sigma \circ \tau$  and  $\tau \circ \sigma$  are different, which shows that in the symmetric group  $S_3$ , the composition of permutations is *not commutative*, that is,  $\sigma \circ \tau \neq \tau \circ \sigma$ .

## Common types of permutations in groups(e.g.)

- **Example:** Let  $S = \{1, 2, 3\}$ . The symmetric group of degree 3,  $S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$ , has the following composition table under the operation  $\circ$ :

| $\circ$ | (1)     | (1 2)   | (1 3)   | (2 3)   | (1 2 3) | (1 3 2) |
|---------|---------|---------|---------|---------|---------|---------|
| (1)     | (1)     | (1 2)   | (1 3)   | (2 3)   | (1 2 3) | (1 3 2) |
| (1 2)   | (1 2)   | (1)     | (1 2 3) | (1 3 2) | (1 3)   | (2 3)   |
| (1 3)   | (1 3)   | (1 3 2) | (1)     | (1 2 3) | (2 3)   | (1 2)   |
| (2 3)   | (2 3)   | (1 2 3) | (1 3 2) | (1)     | (1 2)   | (1 3)   |
| (1 2 3) | (1 2 3) | (2 3)   | (1 2)   | (1 3)   | (1 3 2) | (1)     |
| (1 3 2) | (1 3 2) | (1 3)   | (2 3)   | (1 2)   | (1)     | (1 2 3) |

The results of permutation operations can be *verified by checking the table*.

- 9.3.1 Semigroups and Idempotent Elements
- 9.3.2 Groups
- 9.3.3 Rings and Fields
- 9.3.4 Lattices and Boolean Algebras

- Definition and Examples of Rings
- Operational Properties of Rings
- Subrings and Their Characterization
- Ring Homomorphisms
- Integral Domains and Fields

### 9.3.3 Rings and Fields

#### ↳ Group & Rings

|                          | Group                                                                                                                                       | Ring                                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Definition               | A set with a <b>single</b> binary operation that satisfies the properties of closure, associativity, identity element, and inverse element. | A set equipped with <b>two</b> binary operations (addition and multiplication) that satisfy specific properties. |
| Operations               | A single operation (usually <b>addition or multiplication</b> )                                                                             | <b>Addition and Multiplication</b>                                                                               |
| Additive Structure       | Ignore                                                                                                                                      | Form an <b>Abelian group</b> (commutative group)                                                                 |
| Multiplicative Structure | Form a <b>group</b>                                                                                                                         | Form a <b>semigroup</b> , not necessarily a group.                                                               |
| Identity Element         | Requires an <b>identity element</b>                                                                                                         | An additive identity <b>0</b> is required, while the multiplicative identity <b>1</b> is optional.               |
| Commutativity            | <b>Not necessarily commutative</b> ; if it is commutative, it is called an Abelian group.                                                   | <b>Addition is commutative</b> , while multiplication is not necessarily commutative.                            |

|                    | Group                                                                                                                                | Ring                                                                                     |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| <b>Example</b>     | $(\mathbb{Z}, +), (\mathbb{R}^*, \cdot)$                                                                                             | $(\mathbb{Z}), (\mathbb{R}[x])$                                                          |
| <b>Goal</b>        | Study of symmetry and transformations, abstract algebraic structures, and representation theory.                                     | Study of structures with two operations, polynomial theory, and algebraic number theory. |
| <b>Application</b> | Physics (symmetry operations), Chemistry (molecular symmetry), Cryptography (encryption algorithms), Computer Science (graph theory) | Linear Algebra (matrix rings), Coding Theory, Computer Algebra, Quantum Mechanics        |



## ■ Understanding Symmetry in Groups:

- (1) *Symmetry* refers to the property of an object remaining unchanged under certain operations. These operations can include rotations, reflections, translations, etc.
- (2) *Group symmetry* refers to the invariance of an object under a set of operations that **satisfy the four fundamental properties** of a group: closure, associativity, existence of an identity element, and existence of inverse elements.
- (3) The *symmetry of cyclic groups* (a special type of group) is reflected in structures with periodic repetition.
- (4) Each *permutation in the symmetric group*  $S_n$  (a special group) can be viewed as a **symmetric transformation** of the positions of elements.

- **Example:** The **symmetry group  $D_3$**  of an equilateral triangle with vertices  $A$ ,  $B$ , and  $C$  includes the following transformations that illustrate group symmetry:
  - (1) A 120-degree clockwise rotation corresponds to the **cyclic permutation  $(ABC)$**  in group  $D_3$  .
  - (2) A 240-degree clockwise rotation corresponds to the **cyclic permutation  $(ACB)$**  in group  $D_3$  .
  - (3) A reflection across the line connecting vertex  $A$  and the midpoint of the opposite side corresponds to the **transposition  $(BC)$**  in group  $D_3$  .
  - (4) A reflection across the line connecting vertex  $B$  and the midpoint of the opposite side corresponds to the **transposition  $(AC)$**  in group  $D_3$  .
  - (5) A reflection across the line connecting vertex  $C$  and the midpoint of the opposite side corresponds to the **transposition  $(AB)$**  in group  $D_3$  .

- **Definition 9.21:** Let  $\langle R, +, \cdot \rangle$  be an algebraic system, where  $R$  is a set, and  $+$  and  $\cdot$  are binary operations. If the following conditions are satisfied:
  - (1)  $\langle R, + \rangle$  forms an Abelian group (commutative group) ,
  - (2)  $\langle R, \cdot \rangle$  forms a semigroup,
  - (3) Multiplication ( $\cdot$ ) is distributive over addition ( $+$ ),then  $\langle R, +, \cdot \rangle$  is called a *ring*.

## ■ Examples:

- (1) The sets of integers, rational numbers, real numbers, and complex numbers form rings under the usual addition (+) and multiplication ( $\cdot$ ). They are called the *ring of integers*  $\mathbb{Z}$ , the *ring of rational numbers*  $\mathbb{Q}$ , the *ring of real numbers*  $\mathbb{R}$ , and the *ring of complex numbers*  $\mathbb{C}$ , respectively.
- (2) The set  $M_n(\mathbb{R})$  of all  $n \times n$  real matrices (with  $n \geq 2$ ) forms a ring under matrix addition and matrix multiplication, called the *ring of  $n \times n$  real matrices*.
- (3) Let  $Z_n = \{0, 1, \dots, n-1\}$ , where  $\oplus$  and  $\otimes$  denote addition and multiplication modulo  $n$ , respectively. Then  $\langle Z_n, \oplus, \otimes \rangle$  forms a ring called the *ring of integers modulo  $n$* .
- (4) The power set  $P(B)$  of a set  $B$  forms a ring under the symmetric difference operation  $\oplus$  and the intersection operation  $\cap$ .

#### ↳ Commutative, unital, and zero-divisor-free rings

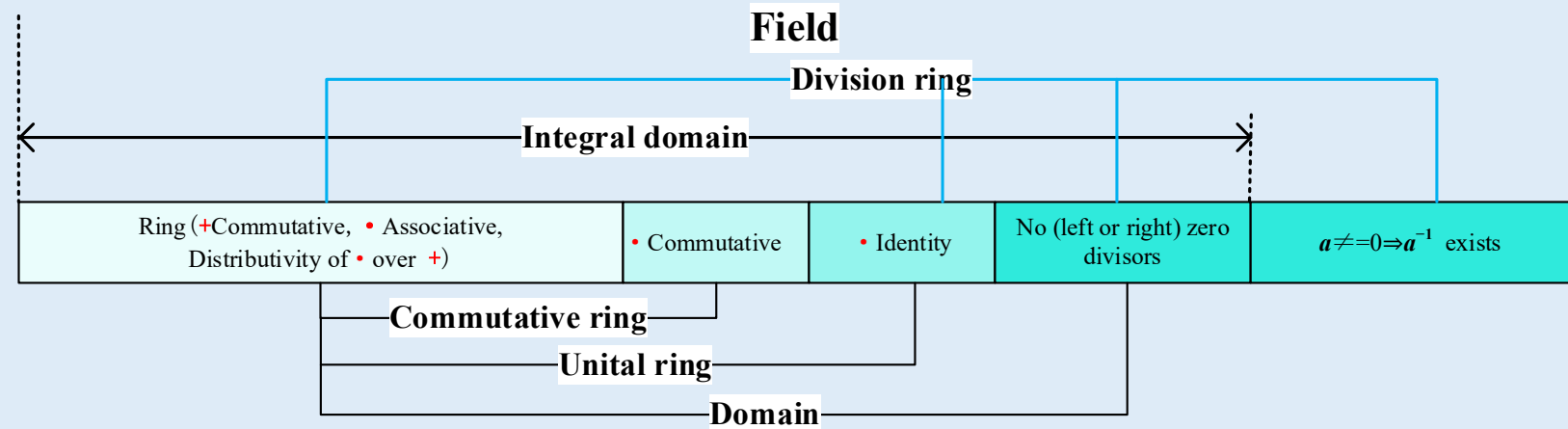
- In a ring  $\langle R, +, \cdot \rangle$ 
  - If multiplication  $(\cdot)$  is commutative, then  $R$  is called a *commutative ring*.
  - If there exists a multiplicative identity for  $\cdot$ , then  $R$  is called a *ring with identity* (or *unit ring*).
  - If there exist elements  $a, b \in R$  such that  $a \neq 0$ ,  $b \neq 0$ , but  $a \cdot b = 0$ , then  $a$  is called a *left zero divisor* and  $b$  a *right zero divisor* in  $R$ .
  - A ring that contains no (left or right) zero divisors is called a *domain without zero divisors*, or simply a *domain*.

| Ring                                            | Commutative Ring | Ring with Identity                                   | Integral Domain                                                            |
|-------------------------------------------------|------------------|------------------------------------------------------|----------------------------------------------------------------------------|
| $\langle \mathbb{Z}, +, \cdot \rangle$          | ✓                | multiplicative identity 1                            | ✓                                                                          |
| $\langle \mathbb{Q}, +, \cdot \rangle$          | ✓                | multiplicative identity 1                            | ✓                                                                          |
| $\langle \mathbb{R}, +, \cdot \rangle$          | ✓                | multiplicative identity 1                            | ✓                                                                          |
| $\langle \mathbb{C}, +, \cdot \rangle$          | ✓                | multiplicative identity 1                            | ✓                                                                          |
| $\langle M_n(\mathbb{R}), +, \cdot \rangle$     | No               | With multiplicative identity:<br>the identity matrix | There exist nonzero<br>matrices $A$ and $B$ such<br>that $A \cdot B = 0$ . |
| $\langle \mathbb{Z}_n, \oplus, \otimes \rangle$ | ✓                | multiplicative identity 1                            | $\mathbb{Z}_6$ has zero divisors,<br>since $2 \otimes 3 = 0$ .             |

## Commutative, unital, and zero-divisor-free rings

### Definition 9.22:

- (1) If the ring  $\langle R, +, \cdot \rangle$  is a commutative ring, has a multiplicative identity, and contains no zero divisors, then  $R$  is called an **integral domain**.
- (2) If the ring  $\langle R, +, \cdot \rangle$  has at least two elements, is a ring with identity and has no zero divisors, and for every  $a \in R$  with  $a \neq 0$ , there exists  $a^{-1} \in R$ , then  $R$  is called a **division ring**.
- (3) If the ring  $\langle R, +, \cdot \rangle$  is both an integral domain and a division ring, then  $R$  is called a **field**.



## Examples: Domains, Division Rings, and Fields

| Ring                                            | Commutative      | Has Unity               | No Zero Divisors                                          | Integral Domain              | Division Ring                                  | Field                        |
|-------------------------------------------------|------------------|-------------------------|-----------------------------------------------------------|------------------------------|------------------------------------------------|------------------------------|
| $\langle \mathbb{Z}, +, \cdot \rangle$          | ✓                | Unity = 1               | ✓                                                         | ✓                            | ✗ Only $\pm 1$ invertible                      | ✗                            |
| $\langle \mathbb{Q}, +, \cdot \rangle$          | ✓                | Unity = 1               | ✓                                                         | ✓                            | ✓ All nonzero elements invertible              | ✓                            |
| $\langle \mathbb{R}, +, \cdot \rangle$          | ✓                | Unity = 1               | ✓                                                         | ✓                            | ✓                                              | ✓                            |
| $\langle \mathbb{C}, +, \cdot \rangle$          | ✓                | Unity = 1               | ✓                                                         | ✓                            | ✓                                              | ✓                            |
| $\langle M_n(\mathbb{R}), +, \cdot \rangle$     | • Noncommutative | Unity = Identity Matrix | ✗<br>$\exists A, B \neq 0, A \cdot B = 0$                 | ✗                            | ✗ nonzero matrix has no multiplicative inverse | ✗                            |
| $\langle \mathbb{Z}_n, \oplus, \otimes \rangle$ | ✓                | Unity = 1               | $\mathbb{Z}_6$ has zero divisors, since $2 \otimes 3 = 0$ | If $n$ prime $\rightarrow$ ✓ | If $n$ prime $\rightarrow$ ✓                   | If $n$ prime $\rightarrow$ ✓ |



## Basic Identities in a Ring (Basic Properties of Rings)

■ Theorem 9.10: Let  $\langle R, +, \cdot \rangle$  be a ring, then

(1)  $\forall a \in R, a \cdot 0 = 0 \cdot a = 0$ .

(2)  $\forall a, b \in R, (-a)b = a(-b) = -(ab)$ .

(3)  $\forall a, b \in R, (-a)(-b) = ab$ .

(4)  $\forall a, b, c \in R, a(b-c) = ab-ac, (b-c)a = ba-ca$ .

■ Example: Let  $\langle R, +, \cdot \rangle$  be a ring,  $\forall a, b \in R$ , compute  $(a+b)^3$  and  $(a-b)^2$ .

• Solve:  $(a+b)^3 = (a+b)(a+b)(a+b) = (a^2+ba+ab+b^2)(a+b)$   
 $= a^3+ba^2+aba+b^2a+a^2b+bab+ab^2+b^3$

$$(a-b)^2 = (a-b)(a-b) = a^2-ba-ab-b(-b) = a^2-ba-ab+b^2$$

• When  $R$  is a commutative ring (i.e.,  $ab=ba$ ), it can be further simplified to:

$$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$$

$$(a-b)^2 = a^2-2ab+b^2$$

- 9.3.1 Semigroups and Idempotent Elements
- 9.3.2 Groups
- 9.3.3 Rings and Fields
- 9.3.4 Lattices and Boolean Algebras

- Definition and Construction of Lattices
- Properties of Lattices - Principle of Duality
- Distributive Lattices, Bounded Lattices, Complemented Lattices

| Characteristics            | Set $S$                                                                    | Partially Ordered Set $\langle S, \leq \rangle$                                                                     |
|----------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Definition                 | An <i>unordered</i> set of distinct elements.                              | A set $S$ equipped with a binary relation that is <i>reflexive</i> , <i>antisymmetric</i> , and <i>transitive</i> . |
| Relations between elements | There is <i>no specific order</i> relationship among the elements.         | There is a <i>partial order</i> among the elements, not all elements are comparable.                                |
| Method of Representation   | Represented using $\{ \}$                                                  | Represented by a <i>Hasse diagram</i> or a directed graph.                                                          |
| Operations                 | Union ( $\cup$ ), intersection ( $\cap$ ), difference ( $\setminus$ ), etc | Least upper bound, greatest lower bound, chain, antichain.                                                          |
| Example                    | $\{1,2,3\}$                                                                | $\{a,b,c\}$ equipped with the relations $a \leq b$ , $b \leq c$                                                     |

**Reflexivity:** For all  $a \in S$ ,  $a \leq a$ .

**Antisymmetry:** For all  $a, b \in S$ , if  $a \leq b$  and  $b \leq a$ , then  $a = b$ .

**Transitivity :** For all  $a, b, c \in S$ , if  $a \leq b$  and  $b \leq c$ , then  $a \leq c$ .

## ↳ Partially Ordered Sets (Poset) and Lattices

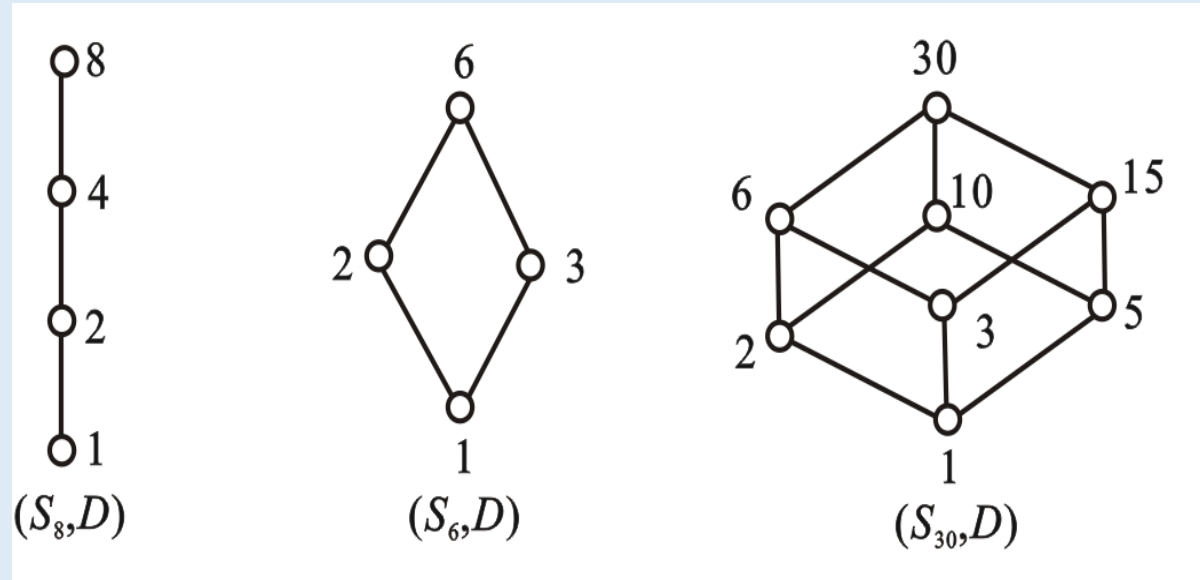
- **Definition 9.23:** Let  $\langle S, \preceq \rangle$  be a partially ordered set. If  $\forall x, y \in S$  the set  $\{x, y\}$  has both a least upper bound and a greatest lower bound, then  $S$  is said to form a *lattice* under the partial order  $\preceq$ .
- Due to the uniqueness of the least upper bound and greatest lower bound, finding the least upper bound and greatest lower bound of  $\{x, y\}$  can be treated as *binary operations*  $\vee$  and  $\wedge$  on  $x$  and  $y$ , where  $x \vee y$  denotes the least upper bound of  $x$  and  $y$ , and  $x \wedge y$  denotes the greatest lower bound of  $x$  and  $y$ .

## ■ Note:

- ① The *binary operations* for least upper bound and greatest lower bound ( $\vee, \wedge$ ) represent operations within the lattice only and have no other meanings.
- ② The lattice  $\langle L, \wedge, \vee \rangle$  constructed from a partially ordered set  $\langle S, \preceq \rangle$  contains not only all elements of  $S$  but also *additional elements* needed to ensure that every pair  $x, y$  has both a least upper bound and a greatest lower bound.
- ③ For any two elements  $x$  and  $y$ , the least upper bound (*supremum*  $x \vee y$ ) is the smallest element among all upper bounds of  $x$  and  $y$ .
- ④ For any two elements  $x$  and  $y$ , the greatest lower bound (*infimum*  $x \wedge y$ ) is the largest element among all lower bounds of  $x$  and  $y$ .

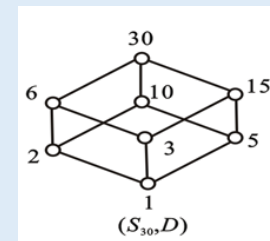
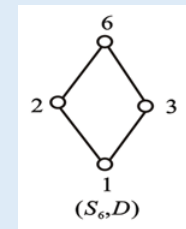
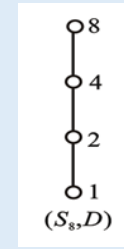
## Partially Ordered Sets (Poset) and Lattices

- **Example:** Let  $n$  be a positive integer, and let  $S_n$  be the set of positive divisors of  $n$ . Let  $D$  denote the divisibility relation. Then the partially ordered set  $\langle S_n, D \rangle$  forms a **lattice**.  $\forall x, y \in S_n$ ,  $x \vee y$  is  $\text{lcm}(x, y)$ , the least common multiple of  $x$  and  $y$ , and the meet  $x \wedge y$  is  $\text{gcd}(x, y)$ , the greatest common divisor of  $x$  and  $y$ .
- Below are the **Hasse diagrams** of the lattices  $\langle S_8, D \rangle$ ,  $\langle S_6, D \rangle$  and  $\langle S_{30}, D \rangle$ .



### ■ Note:

- (1) For the lattice  $\langle S_8, D \rangle$ , the set  $S_8 = \{1, 2, 4, 8\}$  forms a **chain** under the divisibility relation  $D$ , each element divides the element above it, i.e.,  $1 \leq 2 \leq 4 \leq 8$ .
- (2) For the lattice  $\langle S_6, D \rangle$ , the set  $S_6 = \{1, 2, 3, 6\}$  forms a **diamond structure** under  $D$ , both 2 and 3 are direct multiples of 1, and 6 is the common multiple of 2 and 3.
- (3) For the lattice  $\langle S_{30}, D \rangle$ , the set  $S_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$  forms a **cubic structure** under the divisibility relation  $D$ , edges between vertices represent divisibility, e.g., both 2 and 3 divide 6, and both 2 and 5 divide 10.





## Constructing Examples of Lattices

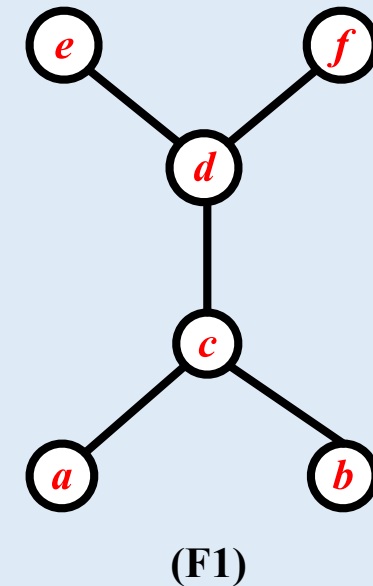
- **Example 1:** Determine whether the partially ordered set  $\langle P(B), \subseteq \rangle$  forms a lattice, where  $P(B)$  is the power set of a set  $B$ .
  - **Solution :**
    - (1) **Check the least upper bound:** For any two subsets  $A, C \in P(B)$ , define their least upper bound as  $A \cup C$ . Under the subset relation, any subset  $D$  that contains both  $A$  and  $C$  must satisfy  $A \cup C \subseteq D$ . Hence,  $A \cup C$  is the least upper bound of  $A$  and  $C$ .
    - (2) **Check the greatest lower bound:** For any two subsets  $A, C \in P(B)$ , define their greatest lower bound as  $A \cap C$ . Under the subset relation, any subset  $E$  that is contained in both  $A$  and  $C$  must satisfy  $E \subseteq A \cap C$ . Thus,  $A \cap C$  is the greatest lower bound of  $A$  and  $C$ .
    - (3) Since for any elements in  $P(B)$ , both the least upper bound  $A \cup C$  and the greatest lower bound  $A \cap C$  exist, the partially ordered set  $\langle P(B), \subseteq \rangle$  **forms a lattice**.

## ↳ Constructing Examples of Lattices

- **Example 2:** Determine whether the partially ordered set  $\langle \mathbb{Z}, \leq \rangle$  forms a lattice, where  $\mathbb{Z}$  is the set of integers and  $\leq$  is the less-than-or-equal-to relation.
  - **Solution:**
    - (1) For any two elements  $a$  and  $b$  in  $\mathbb{Z}$ , the least upper bound is the greater of the two:  $a \vee b = \max(a, b)$ .
    - (2) For any two elements  $a$  and  $b$  in  $\mathbb{Z}$ , the greatest lower bound is the smaller of the two:  $a \wedge b = \min(a, b)$ .
    - (3) Since these conditions hold for all pairs of integers  $a$  and  $b$ , the partially ordered set  $\langle \mathbb{Z}, \leq \rangle$  *satisfies the definition of a lattice*.

## Constructing Examples of Lattices

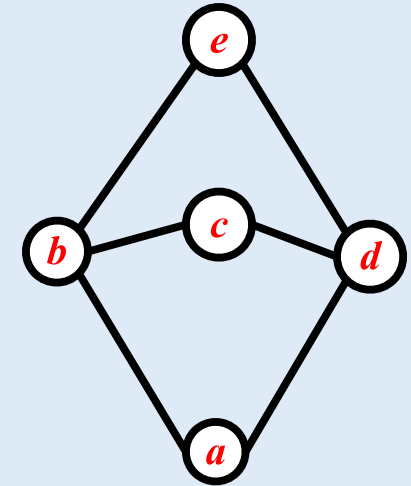
- **Example 3:** Determine whether the structure forms a lattice based on the Hasse diagram of the partially ordered set.
- **Solution:**
  - (1) In diagram (F1), elements  $e$  and  $f$  have no common upper bound, so a least upper bound cannot be determined. Similarly,  $a$  and  $b$  have no common lower bound, so a **greatest lower bound cannot be determined**. Therefore, the structure **does not form a lattice**.



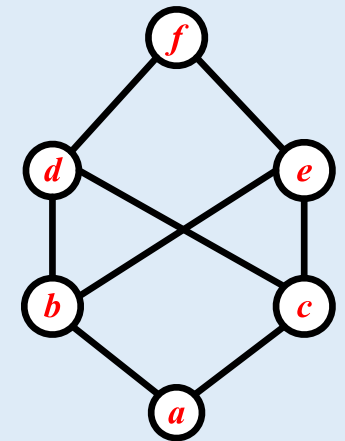
## ■ Solution:

(2) In diagram (F2),  $c$  and  $e$  have no upper bounds, and there is no direct relationship between them. Therefore, a least (common) upper bound greater than both  $c$  and  $e$  cannot be found. The common lower bounds of  $c$  and  $e$  are  $b$ ,  $d$ , and  $a$ . Both  $b$  and  $d$  are greater than  $a$ , and there is no direct partial order between  $b$  and  $d$ . Thus, there is no unique greatest common lower bound. Hence, the structure *does not form a lattice*.

(3) In diagram (F3), the common lower bounds of  $d$  and  $e$  are  $a$ ,  $b$ , and  $c$ . Both  $b$  and  $c$  are greater than  $a$ , and there is no partial order between  $b$  and  $c$ . Therefore, a unique greatest lower bound cannot be determined. As a result, the structure *does not form a lattice*.



(F2)



(F3)

## ↳ Dual statement & Principle of Duality in Lattices

- Let  $f$  be a statement involving elements of a lattice and the symbols  $=, \leq, \geq$ . Define  $f^*$  as the statement obtained by replacing  $\leq$  with  $\geq$ ,  $\geq$  with  $\leq$ ,  $\vee$  with  $\wedge$ , and  $\wedge$  with  $\vee$  in  $f$ . The statement  $f^*$  is called the **dual** of  $f$ .
- **Principle of Duality for Lattices:**  
If a statement  $f$  is true for all lattices, then its dual statement  $f^*$  is also true for all lattices.
- **Example :** In a lattice, if the statement  $f : (a \vee b) \wedge c \leq c$  holds, then its dual statement  $f^* : (a \wedge b) \vee c \geq c$  also holds.

- Theorem 9.11 : Let  $\mathbf{v}$  be a lattice. Then the operations  $\vee$  and  $\wedge$  satisfy the *commutative*, *associative*, *idempotent*, and *absorption* laws, namely:

(1)  $\forall a, b \in L, a \vee b = b \vee a$  and  $a \wedge b = b \wedge a$

(2)  $\forall a, b, c \in L, (a \vee b) \vee c = a \vee (b \vee c)$  and  
 $(a \wedge b) \wedge c = a \wedge (b \wedge c)$

(3)  $\forall a \in L, a \vee a = a$  and  $a \wedge a = a$

(4)  $\forall a, b \in L, a \vee (a \wedge b) = a$  and  $a \wedge (a \vee b) = a$

#### ■ Proof(1) : Commutative Law

- $a \vee b$  is the least upper bound of the set ,  $b \vee a$  is the least upper bound of the set  $\{b, a\}$ .
- Since  $\{a, b\} = \{b, a\}$  we have  $a \vee b = b \vee a$ .
- By the principle of duality, it follows that  $a \wedge b = b \wedge a$  .
- Thus, the commutative law holds.

## ↳ Fundamental Properties of Lattice Operations

### ■ Proof(2) : Associative Law

- From the definition of the least upper bound, we have the following inequalities:

$$(a \vee b) \vee c \geq a \vee b \geq a \quad \textcircled{1}$$

$$(a \vee b) \vee c \geq a \vee b \geq b \quad \textcircled{2}$$

$$(a \vee b) \vee c \geq c \quad \textcircled{3}$$

- From  $\textcircled{2}$  and  $\textcircled{3}$

$$(a \vee b) \vee c \geq b \vee c \quad \textcircled{4}$$

- Combining  $\textcircled{1}$  and  $\textcircled{4}$  we obtain  $(a \vee b) \vee c \geq a \vee (b \vee c)$ .
- Similarly, we can prove:  $(a \vee b) \vee c \leq a \vee (b \vee c)$ .
- By the **antisymmetry** of the partial order, it follows that:  
 $(a \vee b) \vee c = a \vee (b \vee c)$ .
- By the principle of duality,  $(a \wedge b) \wedge c = a \wedge (b \wedge c)$  can also be proved, that is, the **associative law holds**.



### ↳ Definition of a Lattice - Algebraic Perspective

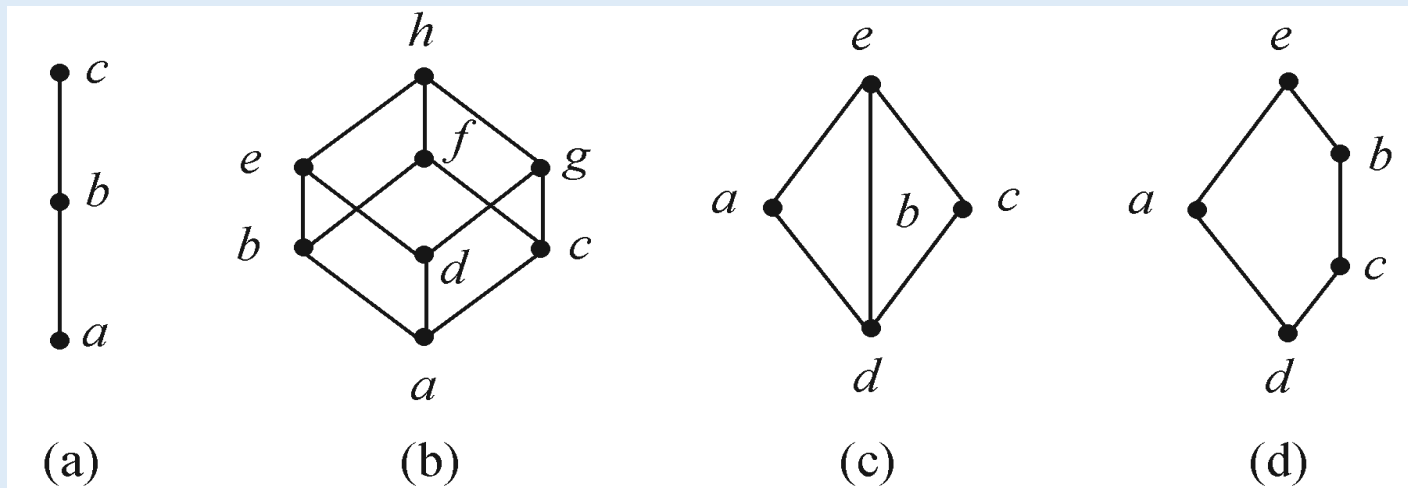
- A **lattice** can be *characterized as an algebraic structure* with join ( $\vee$ ) and meet ( $\wedge$ ) operations satisfying commutativity, associativity, and absorption laws, *or as a poset* in which every pair of elements has a least upper bound and a greatest lower bound.
- **Definition 9.24 :** Let  $(S, *, \circ)$  be an **algebraic system** with two binary operations. If the operations  $*$  and  $\circ$  satisfy the **commutative law**, **associative law**, and **absorption law**, then a partial order  $\leq$  can be suitably defined on  $S$  such that  $\langle S, \leq \rangle$  forms a *lattice*, and for all  $a, b \in S$ , we have:  $\forall a, b \in S, a \wedge b = a * b$  and  $a \vee b = a \circ b$ .
- **Example :** Verify that the algebraic definition of a lattice and the order-theoretic definition are equivalent in characterizing the same lattice structure.

## ↳ Definition of a Lattice - Algebraic Perspective(e.g.)

- ① Let  $S=\{0,1,2,3\}$ , for any  $a,b \in S$ , **define**  $a*b=\min(a,b)$ ,  $a \circ b=\max(a,b)$ .
- ② Define the **partial order**  $\leq$  by  $a \leq b$ , if and only if  $a*b=\min(a,b)=a$  or  $a \circ b=\max(a,b)=b$ .
- ③ **Reflexivity**:  $a*a=\min(a,a)=a$ ,  $a \circ a=\max(a,a)=a$ , so reflexivity holds.
- ④ **Antisymmetry**:  $a \leq b$  means  $a*b=a$  ( $b*a=b$ ),  $*$  satisfy commutative  $a*b=b*a$ , obtain  $a=b$ , thus, antisymmetry holds.
- ⑤ **Transitivity**:  $a \leq b$  ( $a*b=a$ ),  $b \leq c$  ( $b*c=b$ ), then using the associativity of  $*$ ,  $a*c=a*(b*c)=(a*b)*c=a*c=a$ , transitivity holds.
- ⑥ Define the **greatest lower bound**  $a \wedge b=a*b=\min(a,b)$ , for any  $c \in S$ , if  $c \leq a$  and  $c \leq b$  ( $c*a=c$  and  $c*b=c$ ),  $c*(a*b)=c$ , then  $c \leq (a*b)$ , Hence, a greatest lower bound exists for any pair  $a, b$ .
- ⑦ Define the **least upper bound**  $a \vee b=a \circ b=\max(a,b)$ , for any  $c \in S$ , if  $a \leq c$  and  $b \leq c$  ( $a \circ c=c$  and  $b \circ c=c$ ),  $(a \circ b) \circ c=c$ , then  $(a \circ b) \leq c$ , Hence, a least upper bound exists for any pair  $a, b$ .

↳ Distributive lattice = lattice + distributive law

- **Definition 9.25:** Let  $\langle L, \wedge, \vee \rangle$  be a lattice. If for all  $\forall a, b, c \in L$  we have  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ ,  $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$ , then  $L$  is called a *distributive lattice*.
- **Example:** Identify which lattices in the diagram are distributive lattices.

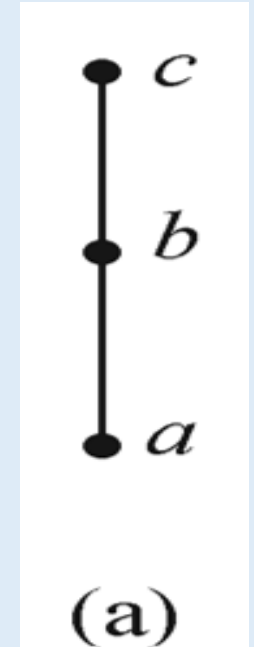


**Solution:** (a) and (b) are distributive lattices, (c) and (d) are not.

## ↳ Examples of Determining Distributive Lattices

- **Solution:** Determine whether lattice (a) is a distributive lattice.

- ① The three elements  $a, b, c$  in the diagram form a simple chain  $a \leq b \leq c$ , where every pair of elements has a well-defined least upper bound and greatest lower bound.
- ② **Verify Distributive Law 1:** Since  $b \leq c$ , we have  $b \vee c = c$ , so:  
 $a \wedge (b \vee c) = a \wedge c = a$ . Since  $a \leq b \leq c$ , we have  $a \wedge b = a$ ,  $a \wedge c = a$ , so  
 $(a \wedge b) \vee (a \wedge c) = a \vee a = a$ . The identity  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$  holds.
- ③ **Verify Distributive Law 2:** Since  $b \leq c$ , we have  $b \wedge c = b$ , so  
 $a \vee (b \wedge c) = a \vee b = b$ . Since  $a \leq b \leq c$ , we have  $a \vee b = b$ ,  $a \vee c = c$ , so  
 $(a \vee b) \wedge (a \vee c) = b \wedge c = b$ . Thus, the identity  $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$  holds.



Therefore, lattice (a) *satisfies both distributive laws* and is a distributive lattice.

## Examples of Determining Distributive Lattices

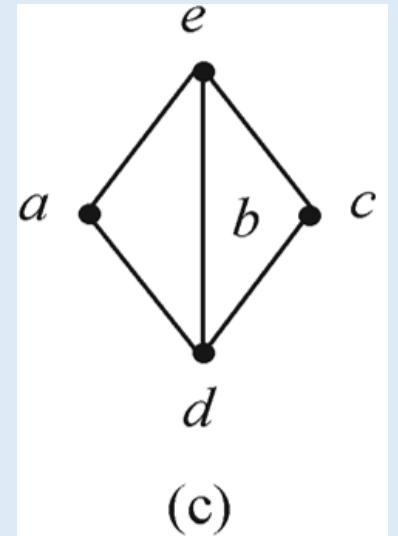
■ **Solution:** Determine whether lattice (c) is a distributive lattice.

① The elements  $a, b, c, d, e$  satisfy the relations:  $d \leq a, d \leq c, a \leq e, c \leq e, d \leq b, b \leq e$ . Each pair of elements has a well-defined least upper bound and greatest lower bound.

② **Verify Distributive Law 1:** Since  $b \vee c = e, a \wedge e = a$ , we get  $a \wedge (b \vee c) = a \wedge e = a$ . Since  $a \wedge b = d, a \wedge c = d$ , we have  $(a \wedge b) \vee (a \wedge c) = d \vee d = d$ . Results:  $a \wedge (b \vee c) \neq (a \wedge b) \vee (a \wedge c)$ , so **Distributive Law 1 does not hold.**

③ **Verify Distributive Law 2:** Since  $b \wedge c = d, a \vee d = a$ , we have  $a \vee (b \wedge c) = a \vee d = a$ . Since  $a \vee b = e, a \vee c = e$ , we have  $(a \vee b) \wedge (a \vee c) = e \wedge e = e$ . Results:  $a \vee (b \wedge c) \neq (a \vee b) \wedge (a \vee c)$ , so **Distributive Law 2 also does not hold.**

Since neither distributive law holds, **lattice (c) is not a distributive lattice.**

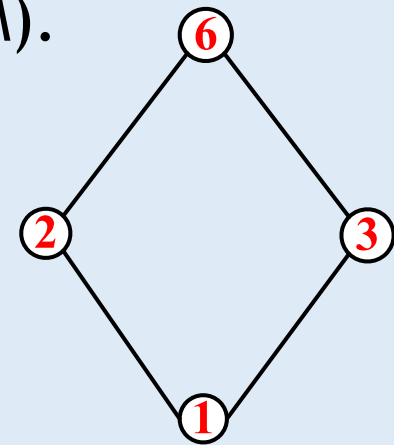


↳ Bounded lattice=lattice+greatest lower bound **0** +least upper bound **1**

- **Definition 9.26** : If there exists an element  $a$  in a lattice  $\langle L, \wedge, \vee, \rangle$  such that for  $\forall b \in L$ ,  $a \preceq b$  (or  $b \preceq a$ ), then  $a$  is called the **greatest lower bound** (or **least upper bound**) of  $L$ .
  - If the **greatest lower bound** of  $L$  exists, it is unique and denoted by **0**.
  - If the **least upper bound** of  $L$  exists, it is also unique and denoted by **1**.
  - If both the greatest lower bound and the least upper bound exist in  $L$ , then  $L$  is called a **bounded lattice**, denoted as  $\langle L, \wedge, \vee, 0, 1 \rangle$ .

## Examples of Determining Bounded Lattices

- **Example (1)** : A finite lattice  $L=\{a_1, a_2, \dots, a_n\}$  is a bounded lattice, where  $a_1 \wedge a_2 \wedge \dots \wedge a_n$  is the **greatest lower bound** (bottom element), and  $a_1 \vee a_2 \vee \dots \vee a_n$  is the **least upper bound** (top element) of  $L$ .
- ① Consider the finite lattice  $L=\{1, 2, 3, 6\}$ , with the **divisibility** relation as the partial order. The **meet operation**  $\wedge$  represents the **greatest common divisor (GCD)** of two elements, and the **join operation**  $\vee$  represents the **least common multiple (LCM)**.
- ② The **greatest lower bound (GCD of all elements)**:  
 $1 \wedge 2 \wedge 3 \wedge 6 = 1$  — the GCD of all elements.
- ③ The **least upper bound (LCM of all elements)**:  
 $1 \vee 2 \vee 3 \vee 6 = 6$  — the LCM of all elements.



## Examples of Determining Bounded Lattices

- **Example(2):** The power set lattice  $P(B)$  is a **bounded lattice**, even when  $B$  is an infinite set.
  - ① The empty set  $\emptyset$  is a subset of every set in  $P(B)$ , making it the **greatest lower bound** (bottom element) of all elements.
  - ② The set  $B$  is a superset of every subset in  $P(B)$ , making it the **least upper bound** (top element).
  - ③ Regardless of whether the set  $B$  is finite or infinite, the power set  $P(B)$  always has a well-defined **bottom element** (the empty set  $\emptyset$ ) and **top element** (the set  $B$ ), and thus forms a **bounded lattice**  $\langle P(B), \subseteq, \cap, \cup, \emptyset, B \rangle$ .

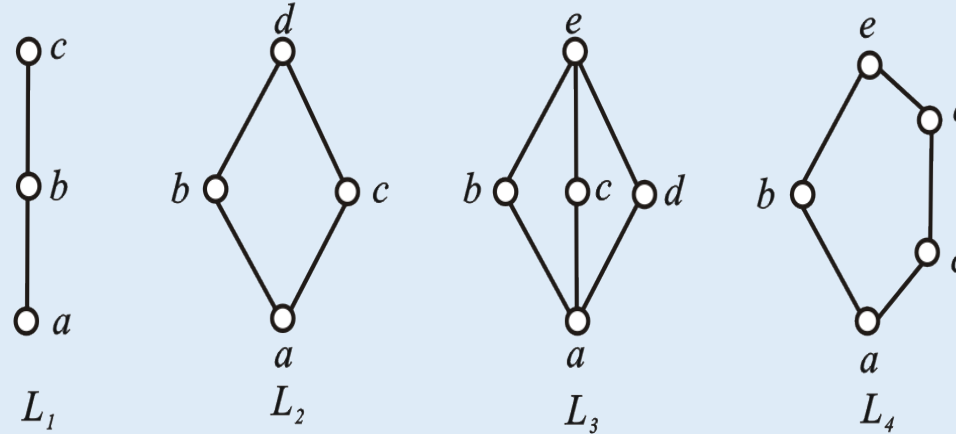


## ↳ Complement element &amp; Complemented lattice

- **Definition 9.27:** Let  $\langle L, \wedge, \vee, 0, 1 \rangle$  be a bounded lattice, and let  $a \in L$ . If there exists  $b \in L$  such that:  $a \wedge b = 0$  and  $a \vee b = 1$ , then  $b$  is called a **complement** of  $a$ .
- The elements **0** (greatest lower bound) and **1** (least upper bound) are **complements of each other**, and each has a **unique** complement.
- Some elements may **not** have any complements, some may have **exactly one**, and some may have **more than one**.
- If **every element** in the bounded lattice  $\langle L, \wedge, \vee, 0, 1 \rangle$  **has a complement**, then the lattice is called a **complemented lattice**.

## Examples of Determining Complemented Lattices

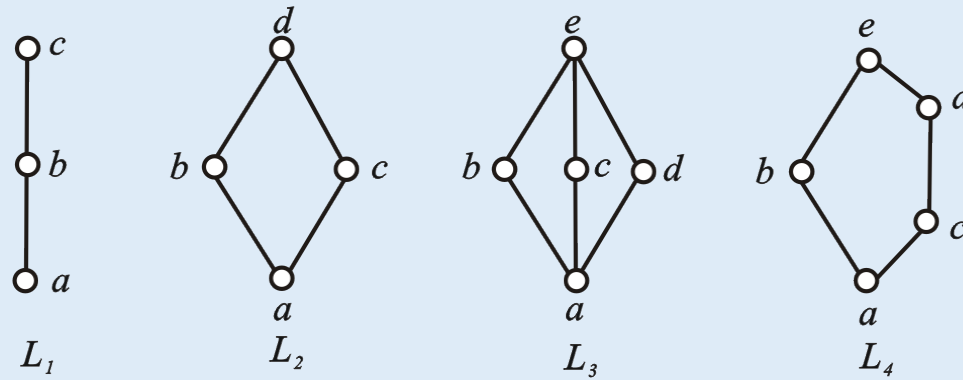
- **Example:** Find the complements of all elements in each of the following four lattices, and determine whether each lattice is a complemented lattice.



• **Solution:**

- (1) In  $L_1$ ,  $a \wedge c = a$  and  $a \vee c = c$  (the maximum element), so  $a$  and  $c$  are complements of each other. However,  $b$  has **no complement**, so  $L_1$  is not a complemented lattice.
- (2) In  $L_2$ ,  $a$  and  $d$  are complements of each other, and  $b$  and  $c$  are complements of each other. Therefore,  $L_2$  is a **complemented lattice**.

## Examples of Determining Complemented Lattices



- (3) In  $L_3$ ,  $a$  and  $a$  are complements of each other. Element  $b$  has two complements:  $c$ ,  $d$ . Element  $c$  has complements  $b$  and  $d$ . Element  $d$  has complements  $b$  and  $c$ . Therefore,  $L_3$  is a **complemented lattice**.
- (4) In  $L_4$ ,  $a$  and  $e$  are complements of each other. Element  $b$  has complements  $c$  and  $d$ . Element  $c$  has complement  $b$ . Element  $d$  has complement  $b$ . Therefore,  $L_4$  is a **complemented lattice**.

### ↳ Definition of Boolean Lattice (Boolean Algebra)

- **Definition 9.28:** If a bounded lattice  $\langle L, \wedge, \vee, 0, 1 \rangle$  is both complemented and distributive, then  $L$  is called a **Boolean lattice** or **Boolean algebra**.
- A **distributive lattice** must satisfy the distributive laws, that is, for all  $a, b, c \in L$ :  $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$  and  $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$ .
- A **complemented distributive lattice** not only requires the existence of complements but also that each complement is **unique**.
- In a Boolean lattice (algebra), the operation of taking the **complement** can be regarded as a **unary operation** in Boolean algebra.
- A Boolean algebra is typically denoted as  $\langle B, \wedge, \vee, ', 0, 1 \rangle$ , where  $'$  represents the **complement operation**.

- **Example:** Determine whether the power set lattice  $P(B)$  is a Boolean algebra.
- **Solution:**
  - (1) To determine whether  $P(B)$  is a *lattice*: For any two subsets  $A, B \subseteq B$ , both the union  $A \cup B$  (least upper bound) and the intersection  $A \cap B$  (greatest lower bound) exist and belong to  $P(B)$ , so the definition of a lattice is satisfied.
  - (2) To determine whether  $P(B)$  is a *distributive lattice*: For any three elements in  $P(B)$ , the operations of union  $\cup$  and intersection  $\cap$  satisfy the **distributive laws**, so the definition of a distributive lattice is fulfilled.

- (3) To determine whether  $P(B)$  is a *complemented lattice*: For every subset  $A \subseteq B$ , there exists a complement  $A^c = B - A$  such that  $A \cap A^c = \emptyset$  (the bottom element) and  $A \cup A^c = B$  (the top element), satisfying the definition of a complemented lattice.
- (4) To determine whether  $P(B)$  is a *Boolean lattice*: Since  $P(B)$  is already known to be a lattice, a distributive lattice, and a complemented lattice, it follows that the power set lattice  $P(B)$  is a Boolean lattice (Boolean algebra).

- Note:

The *uniqueness of complements* in the Boolean lattice  $P(B)$  can be proven using the properties of Boolean algebra — namely, the commutative, associative, and distributive laws.

## Structure Theorem for Finite Boolean Algebras

- Theorem 9.12: Let  $L$  be a finite Boolean algebra. Then  $L$  contains  $2^n$  elements for some  $n \in \mathbb{N}$ , and  $L$  is *isomorphic to the Boolean algebra*  $\langle P(S), \cap, \cup, \sim, \emptyset, S \rangle$ , where  $S$  is a set with  $n$  elements.
- Note:
  - ① An *isomorphism* between two Boolean algebras *requires* a **bijection** that preserves **meet** ( $\wedge$ ), **join** ( $\vee$ ), and **complement** operations, as well as the **zero** and **unit** elements.
  - ② Up to isomorphism, there is *only one Boolean algebra* with  $2^n$  elements.

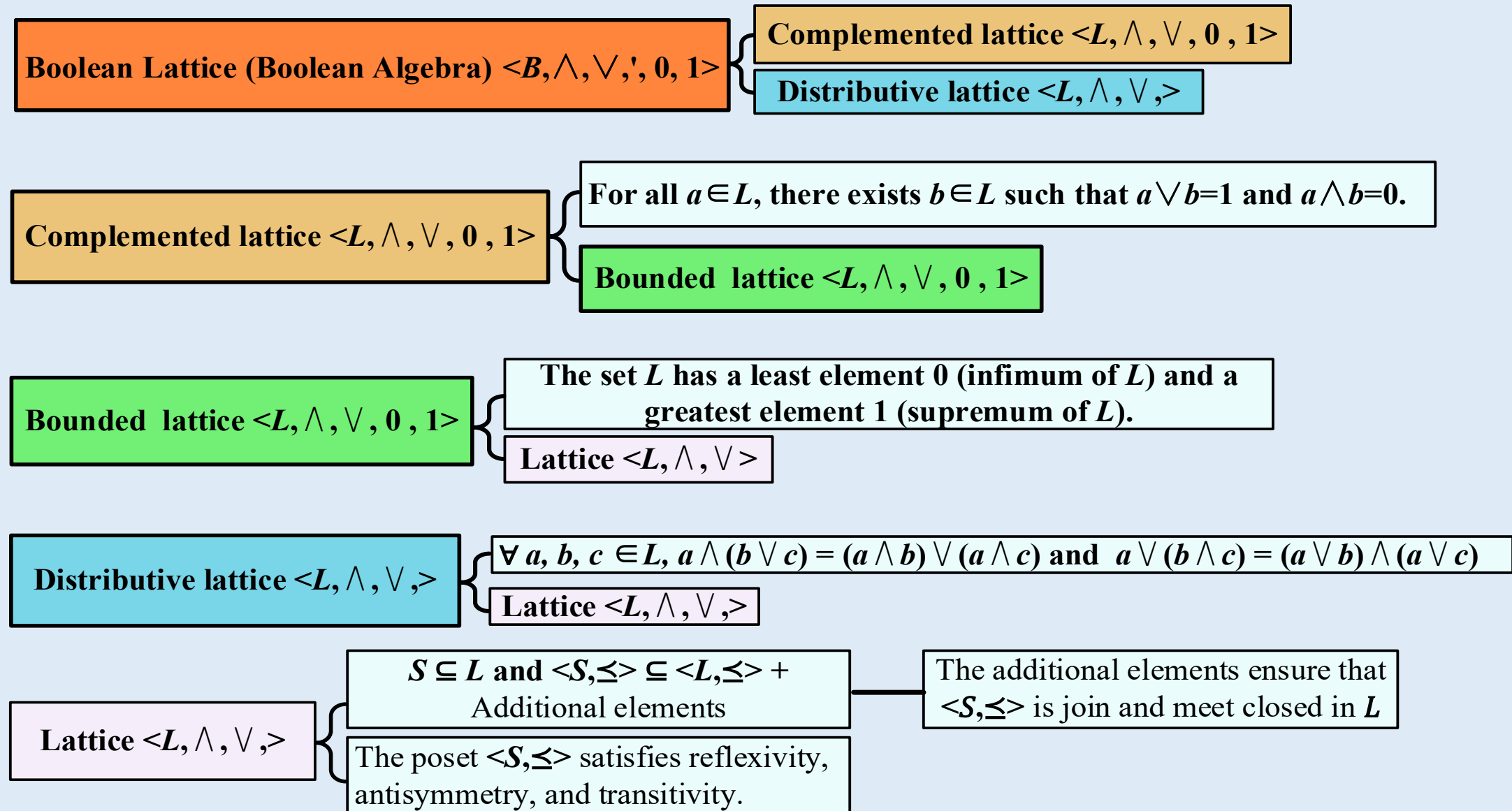
## ↳ Structure Theorem for Finite Boolean Algebras

### ■ Note:

- ③ Any finite Boolean algebra  $L$  is isomorphic to the power set algebra  $\langle P(S), \cap, \cup, \sim, \emptyset, S \rangle$ , where  $P(S)$  is the power set of an  $n$ -element set  $S$ . This *intuitive model* facilitates a better understanding and manipulation of Boolean algebra.
- ④ This theorem *bridges* abstract algebra (Boolean algebras) and set theory (power set algebras), enabling broad applications in **logic**, **computer science**, and **algebraic geometry**.
- ⑤ Understanding the *algebraic structure* and its isomorphism to *set-theoretic models* aids in simplifying logical expressions and optimizing algorithm design.



# From lattices to Boolean algebra



## 9.3 Several Typical Algebraic Systems • Brief summary

**Objective :**

**Key Concepts :**

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**Key Concepts :**